Double Pendulum Arm Kit Installation Manual

This manual provides instructions for the mechanical assembly, electrical connections, and workbench mounting of the Double Pendulum Arm Kit. This product is intended only for users 18 years of age or older and solely for educational, experimental, and technical demonstration purposes.

Safety Instructions

WARNING This product has moving pendulum arms, pinch points, and 48 VDC electrical connections. Incorrect assembly, wiring, or operation could result in serious personal injury, electric shock, short circuit, fire, or property damage. Read and follow this manual in full before assembly or operation.

- This product is not a toy. Persons under 18 years of age must not assemble, wire, commission, or operate it.
- Disconnect and isolate all power before assembly, wiring, adjustment, cleaning, or inspection. Do not touch moving parts until they have come to a complete stop.
- Use only a 48 VDC power supply that meets the product requirements. Before energizing, verify polarity, terminals, pins, and wire connections.
- Before each use, inspect all screws, nuts, arms, the encoder, cables, and C-clamps. Do not operate with loose, damaged, or missing parts.
- Keep hands, body, hair, loose clothing, jewelry, and tools away from the arms, joints, and all moving parts. Wear suitable eye protection during setup and operation.
- Maintain a clear safety zone around the full range of arm motion. Remove tools and obstructions, keep bystanders away, and never leave the product operating unattended.
- Install only on a level, rigid, stable work surface. Before energizing, confirm that the C-clamps cannot slip or loosen.
- Immediately disconnect power and stop use if you observe looseness, abnormal vibration or noise, overheating, odor, sparks, cable damage, or control malfunction.
- Do not make unauthorized modifications to the mechanical structure, electrical wiring, control system, or safety-related parts. Do not use damaged or incompatible components.
- Do not use the product in damp, wet, flammable, explosive, or unstable environments.

Disclaimer

This product is intended solely for adult educational, experimental, and technical demonstration use. It is not intended for industrial production, safety-critical systems, or unattended operation. The user is responsible for installing and operating the product in accordance with this manual and all applicable local laws, codes, and safety requirements, and for confirming that the operating environment and associated equipment are suitable.

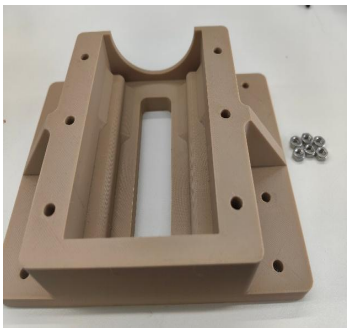
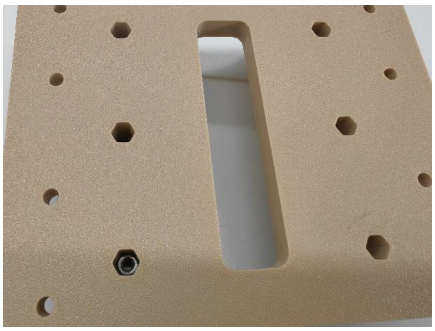
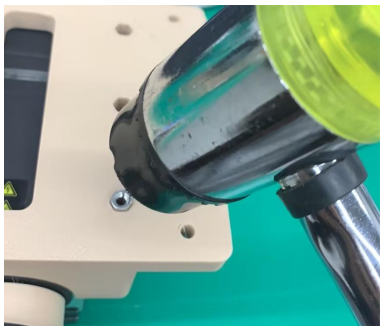
To the maximum extent permitted by applicable law, the manufacturer shall not be liable for indirect, incidental, or consequential damages arising from misuse, incorrect wiring, an unsuitable power supply, insecure mounting, unauthorized modification, use outside the intended purpose, or failure to follow this manual. The manufacturer makes no representation that the product is suitable for any specific application not confirmed in writing.

Nothing in this disclaimer excludes or limits any consumer right, liability for a defective product, liability for personal injury, or other liability that cannot lawfully be excluded or limited. Any warranty is governed solely by separate written warranty terms issued by the manufacturer.

Base and Motor Installation

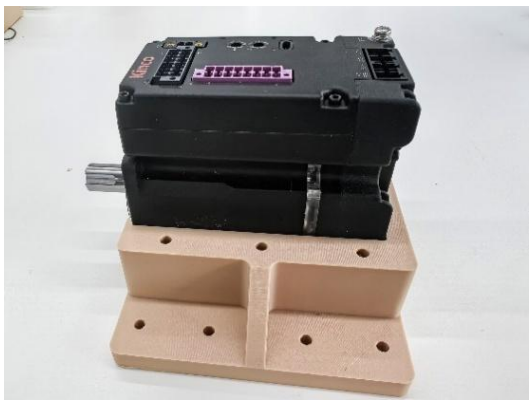
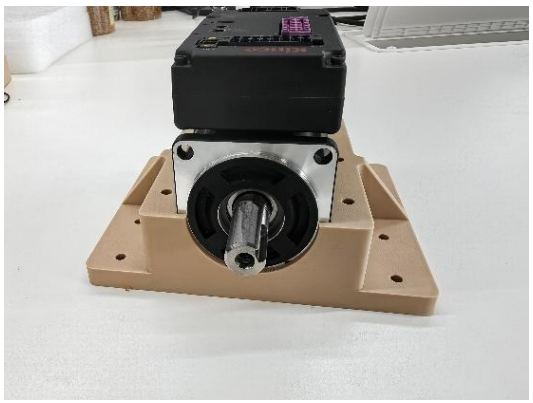
WARNING Keep all power disconnected during assembly and wiring. Do not energize the product until all mechanical fasteners are secure and all wiring has been checked.

Step 1 Take out the six M4 nuts and the 3D-printed motor mounting base. Insert the nuts into the recesses on the underside of the base. Gently tap the nuts into the recesses with a hammer. Then, use a screwdriver to press against the nut and tap the screwdriver handle with the hammer until the nut reaches the bottom of the recess.

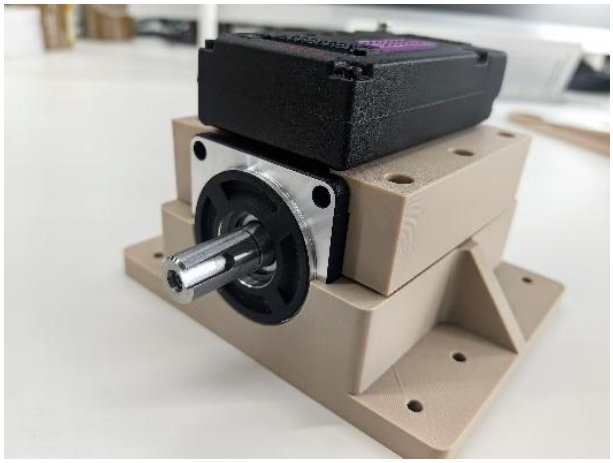






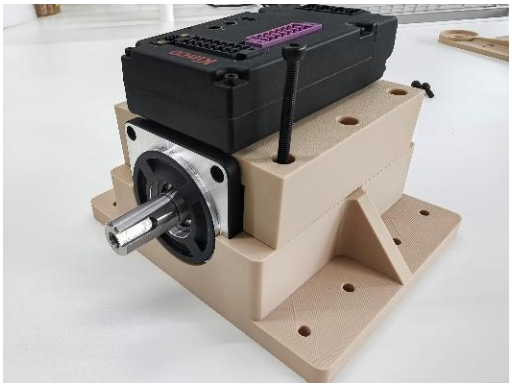
Step 2 Position the motor on the motor mounting base.

Step 3 Install one motor end clamping plate at each end of the motor and mounting base.



Step 4 Secure the clamping plates to the mounting base with six M4 screws. The M4 screw is designed with minimal margin and no excess length. Ensure the nut from Step 1 rests fully at the bottom of the recess, and the M4 screw is fully tightened to engage with the nut. Otherwise, the clamping plate will not be adequately clamped and may work loose. To ensure a secure clamp, turn the motor base on its side. Use a screwdriver to hold the nut firmly at the bottom of the recess while tightening the M4 screw from the other side.



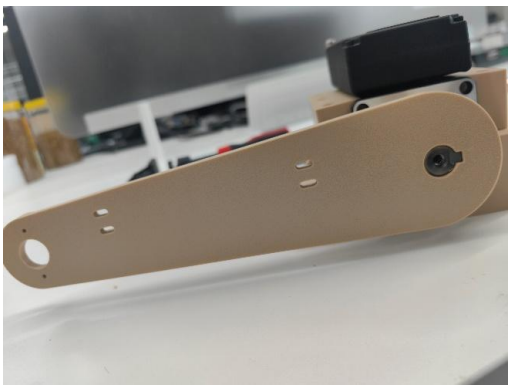


Drive Arm and Encoder Installation

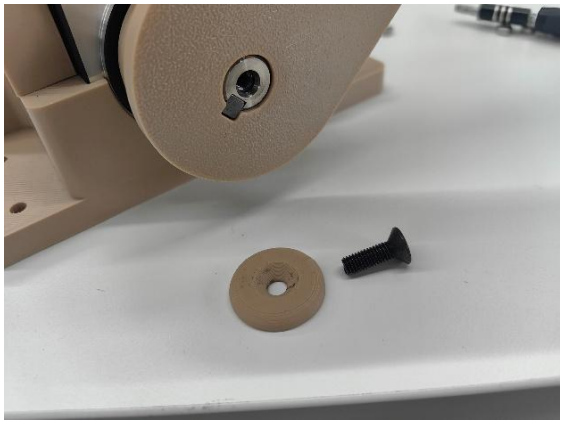
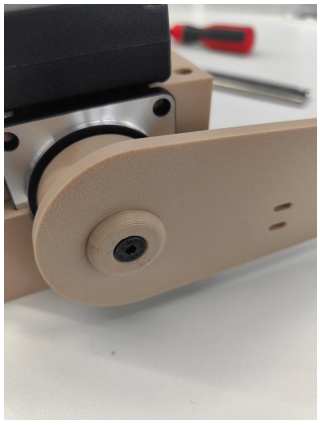
Step 5 Insert the flat key into the keyway of the motor shaft, then press it in with a C-clamp to ensure the flat key fully seats against the bottom of the keyway. Align the bore on the drive arm with the motor shaft and press the drive arm axially by hand until fully seated. If hand pressing alone is difficult, apply a sharp axial pat with the center of the palm to fully seat the drive arm via instantaneous impact.







Step 6 Install the 3D-printed circular retainer on the drive arm bearing and secure it with the M5 x 16 countersunk screw.

Step 7 Remove the sheet-metal mounting bracket from the encoder using a screwdriver. (Note: Use a properly sized screwdriver that can effectively transmit torque. Using an incorrect size may slip and strip the screw head.) Loosen the set screw, insert the dowel pin into the incremental encoder, then tighten the set screw.



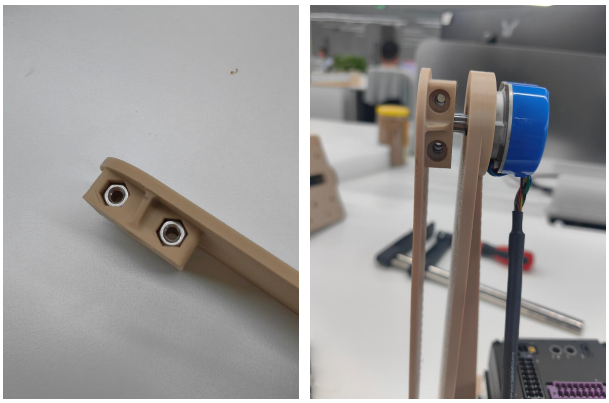


Step 8 Secure the incremental encoder to the drive arm with two M2 screws.

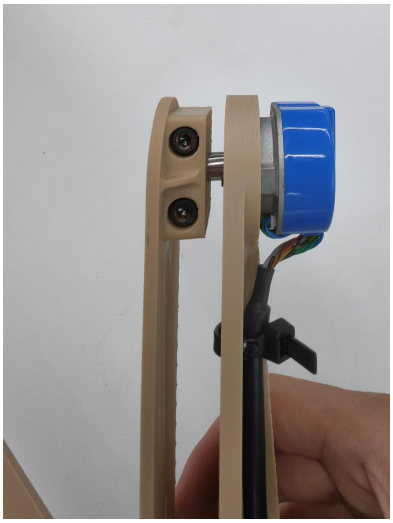


CAUTION Pinch points exist at the arm joints. Support both arms during assembly, maintain clearance between them, and keep fingers out of the gap.

Step 9 Insert two M4 nuts into the 3D-printed free pendulum arm. Fit the free pendulum arm over the cylindrical dowel pin on the drive arm, leaving clearance between the two arms.

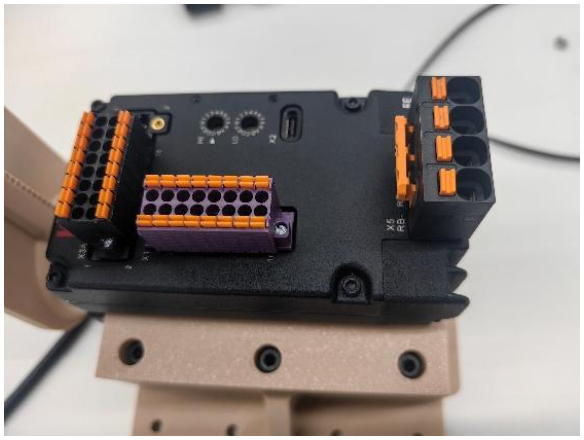


Step 10 Secure the free pendulum arm to the cylindrical dowel pin with two M4 screws.



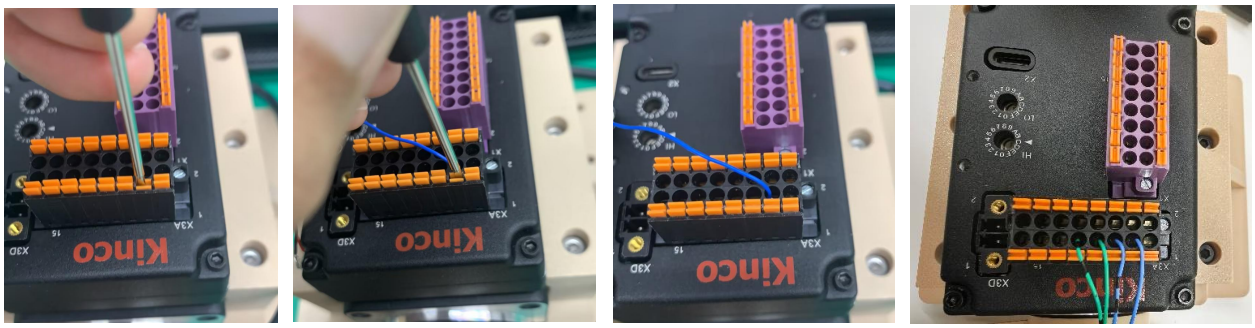
Electrical Connections and Cable Management

Step 11 Install the three motor terminal connectors on the motor.



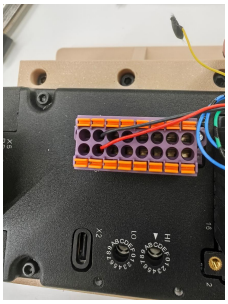
WARNING Disconnect the 48 VDC power supply before wiring. Never wire the product while energized. Incorrect terminals, pins, or polarity could cause electric shock, short circuit, fire, or equipment damage.

Step 12 Connect the encoder pulse-output wires to terminal X3A. Press the release slot down fully with a screwdriver, insert the wire, and release to clamp. Verify each wire color and pin number against the table below.



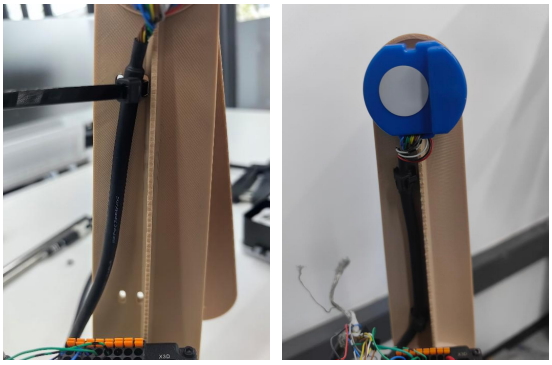
Wire Color	Terminal	Pin
Blue	X3D	3
Blue/Black	X3D	5
Green	X3D	7
Green/Black	X3D	9

Step 13 Connect the encoder power wires to terminal X1. Verify each wire color and pin number against the table below.



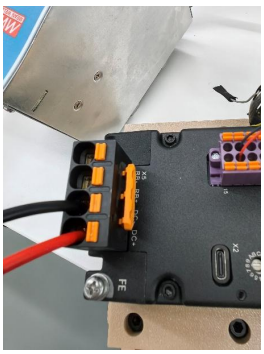
Wire Color	Terminal	Pin
Red	X1	13
Black	X1	14

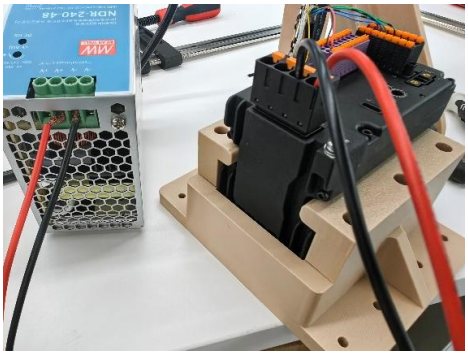
Step 14 Use two cable ties to secure the encoder cable to the free pendulum arm, then trim the excess tie length.



WARNING Before energizing, verify that DC+ is connected to V+ and DC- to V-. When making the connection, carefully align the L, N and PE terminals of the power supply with the corresponding pins on the power inlet to prevent miswiring. Immediately disconnect power if you observe sparks, odor, overheating, or any other abnormal condition.

Step 15 Connect the motor power wires to the 48 VDC power supply. Verify the polarity against the table below.



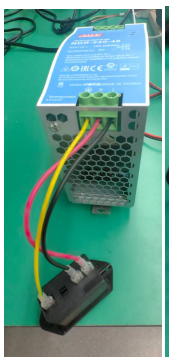


Motor Power Lead	48 VDC Power Supply
DC+	V+
DC-	V-

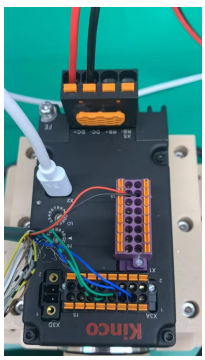
Connect the L, N and PE terminals of the power supply to the corresponding pins on the power inlet (indicated by the red arrows in the figure below). Note that the markings may be indistinct in the image but are clearly visible on the physical unit. Before running the deployment script, turn on the 48V power supply first, then connect the USB Type-C communication cable to port X2, and plug the other end into the USB port on the controller. Otherwise, the motor drive will trigger faults such as logic undervoltage.











Securing the Assembly

WARNING Before energizing, verify that both C-clamps and all fasteners are secure and that no person, tool, or obstruction is within the arms' range of motion. Do not touch the arms during operation or leave the product operating unattended.

Step 16 Secure the assembled Double Pendulum Arm Kit to the workbench or tabletop with the two C-clamps.



Package Contents

Before assembly Verify that all components listed below are present and undamaged. Do not assemble or energize the product if any component is missing, damaged, or incompatible. Contact the supplier for assistance.

Item	Model	Specification	Quantity
Integrated Servo Motor	iSMD60D-040-DQAK-AA-000	400 W; CANopen	1
Incremental Encoder	TS6014N119		1
Free Pendulum Arm	3D-Printed		1
Drive Arm	3D-Printed		1
Motor Mounting Base	3D-Printed		1
Motor End Clamping Plate	3D-Printed		2
3D-Printed Circular Retainer	3D-Printed		1
Cable Tie			2
AC/DC Power Supply	MW NDR-240-48	48 VDC, 5 A	1
Cylindrical Dowel Pin		Ø8 x 35 mm	1
Screw and Nut Kit		M2 x 8 screws (2) M4 nuts (8) M4 x 45 screws (6) M4 x 18 screws (2) M5 x 16 countersunk screw (1)	1 set
C-Clamp		5 in.; 80 x 200 mm	2
Fused IEC C14 Panel-Mount Power Inlet	AC-14-F16C	10 A, 250 VAC	1
USB Type-C Cable		1 m	1
FR PVC Flat Cord	ZR-RVB	2C x 1.5 mm²; 1 m	1
C14-to-C13 Power Jumper Cord		2 m	1
Accessory Kit Packing List			1 copy